

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

We need to understand the concept of data



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain what data is.
- I can give different information related to data.
- I can talk about the importance of data in different fields.

Ask your partner:

- What do you think information stored in computers represents?
- Do you think it is important to store information in computers? Why?





Let's Explore!

The Concept of Data



Instructions: Read the text below.

- Johanna:** Hey, Tom! I'm reading a great book about systems engineering, you know I'm very fond of reading.
- Tom:** Yes, I know. What's the article about?
- Johanna:** It's about the concept of data. Nowadays, we relate data to computing and systems, but centuries ago, scientists and scholars used the term data to refer to a given fact in the world; for instance, a notch in a stick. When we talk about information stored in computers or any digital media, data is a representation of facts from reality. They are characterized as binary states, which are bit patterns with no meaning or sense.
- Tom:** Oh, I heard about that. I usually read an online magazine about systems on bBlog, and the article I'm reading at the moment is about data. They say that data doesn't have meaning by itself, but it needs to be translated by a programming language and defined by data statements. Those tools establish the context of the data, making data meaningful for humans to interpret it. Human beings are the only ones capable of making sense of the data and context. But, what is the use of data in real life?
- Johanna:** Well, data has lots of important usages. For example, in the educational field, software defines a grade as good or bad, as a result of the purpose and interpretation of the test designer. It is not the system the one that decides what grade to deserve, it is the human who programs the system to do so.
I work in computer science, and I'm learning how to understand data through different programming languages; data binary states can be interpreted as letters, numbers, parts of numbers, and so forth. So in the banking field, for instance, data may be translated into clients' codes and figures
- Tom:** That's very fascinating, isn't it?

**Let's Practice!**

Mario, could you explain to me the definition of data?

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

to read / to understand / to clarify

In order to understand the term data, it is necessary _____ about computing science and information systems. For that, I have some written ideas _____ that data represents all information or facts from reality. I also found that only human beings are able _____ that data has a context.

**Let's Practice!**

I am trying to write something about data information.

Instructions: Organize the mystery sentence. Read the sentences aloud. Check your pronunciation with your instructor.

1. say / data / I / to / as / that / / facts. / want / scientists / defined

2. term / a / information / systems. / I / related / / data / is / mention / to / that / to / need

3. that / say / have / only / to / I / data. / interpret / humans



Extension Activity

Let's talk about data.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about a particular field where data can be important.
- Write a short paragraph in which you describe the importance of data in the field you chose.
- Include information about how that field is using data and the purposes of using it.
- Proofread your paragraph.
- Read your paragraph aloud.

Self-Assessment

Yes

No

- I search about field that use data. ☐ ☐
- I describe the importance of data in different fields. ☐ ☐
- I recognize the purposes of using data in a specific field. ☐ ☐



Target Words

- **be fond of** (exp.): to like something very much
e.g. You know I'm very fond of reading.
- **binary state** (noun): the binary data which consist of zeroes and ones
e.g. They are characterized as binary states.
- **figure** (noun): in this case, an amount of money
e.g. So in the banking field, for instance, data may be translated into clients' codes and figures.
- **bit pattern** (noun): the possible combinations (4) among the binary units (zero and one)
e.g. They are characterized as binary states, which are bit patterns with no meaning or sense.
- **meaning** (noun): significance
e.g. Which are bit patterns with no meaning or sense.
- **notch** (noun): a V mark
e.g. Such as a notch in a stick.



Learning Points

Infinitives of Purpose in Tenses with Simple Present.

We can use the **infinitives of verbs** to show the objective of something what we are doing.

The preposition **to** before a verb indicates its **infinitive form**, which is used to express the reason we have for doing something. We can use it at the beginning or in the middle of a sentence.

Examples:

***To** improve the system, we need a new software developer.*

*I check the system in the morning, **to** make sure there are no viruses.*

Remember that the verb with to is the one that shows **the purpose** of the actions.

Examples:

*I work on my software program every day **to finish it on time**.*

Notice that when the preposition **to** is in the middle of the sentence, we do not use a comma to split the sentences.

To talk about the purpose of a habit or a routine we express **our goals** with the preposition to + infinitive, and we use the **simple present** to show the actions taken to achieve our purpose.

Purpose	Actions
<i>To improve the system</i>	<i>we are revising the software.</i>
Actions	Purpose
<i>I always make a back-up copy</i>	<i>to protect the information..</i>

Examples

- *But scientist and scholars, centuries ago, used the term data to refer to a given fact in the world.*
- *Those tools establish the context of the data making data meaningful for humans to interpret data.*
- *Is the human who programs the system to do so.*



Learning Points

Contrast, Simple Present and Present Continuous to Talk about Permanent States, and Current Actions.

We use both the **present simple** and the **present continuous** to talk about the present, but both structures have completely different meanings and uses.

Simple present is a tense used to talk about permanent situations in the present, general facts and truths that are not likely to change.

Examples:

*We **work** at a big multinational.*

*I **know** about information systems.*

The **simple present** has a very simple structure in its affirmative form. But, there is only one important fact you need to remember, when you talk indirectly about one single person or object (third person singular form; **he, she** and **it**) you add an **-s** or an **-es** to the verb.

Example:

*I **work** with software programs.*

*He **knows** a lot about computers.*

On the other hand, we use the **present continuous** tense to talk about temporary situations and about things that are happening at the moment of speaking (or just for a certain period of time) but are going to change in the future.

Examples:

***This year** we're living in Bogotá.*

*I'm studying systems **at the moment**.*

Notice that the **present continuous** is usually accompanied by temporary time expressions such as *at the moment, this week, now* or *right now*.

Do not forget to use the verb **to be** conjugated when you are making a **present continuous** sentence.

Examples:

*Now I **am** working at a systems company.*

*We **are** repairing systems at a café at the moment.*

There are some verbs in English we don't use in the **present continuous**, even if we are talking about a situation happening at the moment of speaking. Those verbs are the ones related to emotions (like, hate), the ones connected to mental functions (think, remember), the verb have, when talking about possessions, and the verb need. These verbs are called **stative verbs**.



Learning Points

Simple present (permanent)	Present continuous (temporary)
• <i>I repair computers.</i>	• <i>I'm repairing computers at the moment.</i>
• <i>We know about systems.</i>	• <i>I know what I'm doing!</i>

Examples

- *I'm reading a great book about systems engineering, you know I'm very fond of reading.*
- *I read a systems online magazine on bBlog every Saturday, the article I'm reading at the moment is about data.*
- *I work in computer science, and I'm learning how to understand data through different programming languages*

List of programming and computer science words.

There are thousands of **words** and **expressions** related to **computer science** and programming that you probably already know. Here we are going to revise some of them.

These are some **terms** and **programs** we use when working on the computer science field:

B2evolution: is a content and community management system supported by a MySQL database. It has GNU public license.

Backup: to copy data, information and others on an extra media, apart from its original.

Baitware: refers to defective software with limited defects, offered for free in order to attract users to commercial products.

bBlog: is a publishing and content system manager used to manage frequently updated information specially from weblogs. It also has a GNU public license.

Bedroom Programming: is a practice which consist of one or a group of amateur programmers developing software (especially computer games) with no intention of making a profit, but just because they love it.

Example

- *I read a systems online magazine on bBlog every Saturday.*

